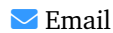


Ghazal Mousavi

[Email](#)[Website](#)[LinkedIn](#)[GitHub](#)[Scholar](#)

Education

University of Tehran: B.Sc. Electrical Engineering

Sept 2021 – Feb 2026

- GPA: 18.35 / 20 (ranked 8th out of 101 students)

University of Tehran: Minor Computer Engineering

Sept 2023 – Feb 2026

- **Relevant Coursework:** Advanced Programming, Discrete Mathematics, Artificial Intelligence, Data Structures and Algorithms

Research Experiences

Research Assistant: University of Alberta

Aug 2026 – Present

Canada – Remote

Supervisor: [Prof. Masoud Ardakani](#)

- Working on Beyond Diagonal Reconfigurable Intelligent Surfaces (BD-RIS) for 6G wireless communications.

Research Assistant: University of Tehran

Mar 2025 – Present

Signal Processing & Communication Systems Laboratory

Tehran, Iran – In person

Supervisor: [Prof. Ali Olfat](#)

- Working on detection algorithms for decentralized massive MIMO uplink systems, with a focus on learning-based approaches.

Researcher: [VirCollab](#)

Jun 2024 – Jun 2025

Canada – Remote

Team Lead: [Dr. Mohammad R. Salmanpour](#)

- Worked with various machine learning evaluation metrics and evaluated model performance across different metrics using MATLAB, Python, and R. Identified discrepancies and developed a standardized library for evaluation metrics.
- Contributed to an image-to-image translation project for ultrasound-to-MRI conversion, gaining hands-on experience with generator and discriminator models.

Research Assistant: [Monash University](#)

Jul 2024 – Jan 2025

Australia – Remote

Supervisor: [Prof. Mehrtash Harandi](#)

- Worked on enhancing machine unlearning through vision transformers for image classification.

Publications

Newton-Based Mini-Batch Detection in Decentralized Large-Scale Antenna

May 2026

Systems (Accepted & Presented at [ICEE 2026](#))

Ghazal Mousavi, Mojtaba Amiri, Ali Olfat

Machine Learning Evaluation Metric Discrepancies Across Programming Languages and Their Components in Medical Imaging Domains: Need for Standardization

Mar 2025

Mohammad R. Salmanpour, Morteza Alizadeh, *Ghazal Mousavi*, Saba Sadeghi, Sajad Amiri, Mehrdad Oveisi, Arman Rahmim, Ilker Hacıhaliloglu

[10.1109/ACCESS.2025.3549702](https://doi.org/10.1109/ACCESS.2025.3549702)

Teaching Assistant Experiences

- | | |
|---|---|
| • Engineering Mathematics
Instructor: Dr. Mahdi Tale Masouleh | • Communication Systems II
Instructor: Prof. Ali Olfat |
| • Signals and Systems
Instructor: Dr. Saeed Akhavan | • Statistics & Probability
Instructor: Dr. Amir Masoud Rabiei |
| • Principles of Communication Systems
Instructor: Prof. Ali Olfat | • Principles of Communication Systems
Instructor: Hamideh Zamanpour |

Projects

Wireless Communication Projects

- Implemented OFDM transmission of 10^7 bits using QPSK modulation over an AWGN channel
- Simulated time diversity over Rayleigh fading channels
- Tools used: MATLAB

Deep Learning and Model Fine-Tuning Projects

- Fine-tuning large language models for Persian conversation using LLaMA 3.2-3B
- Triplet Variational Autoencoder (Tri-VAE) for tumor detection in MRI scans
- Vision Transformer (ViT) for white blood cell classification

Antenna Project

- Designed and simulated a Log-Periodic Dipole Array
- Designed and simulated a Leaky-Wave Antenna based on Substrate Integrated Waveguide ([SIW](#))
- Tools used: Ansys HFSS

Blind Source Separation (BSS) Projects

- Solved Blind Source Separation problems using methods like maximizing kurtosis and minimizing D-Kullback divergence
- Addressed sparse recovery problems, non-stationary and uncorrelated sources, and mutually uncorrelated sources
- Tools used: MATLAB

Skills

Programming Languages: C++, Python, R, Verilog

Tools & Frameworks: MATLAB, TensorFlow, PyTorch, Git, Ansys HFSS, $\text{\LaTeX}$

IELTS: Overall Band Score **7.0** (Listening: 7.5, Reading: 7.0, Writing: 6.5, Speaking: 7.0) [*Advanced Level*]